

SERS-Based Nanobiosensor Workflow for Sepsis Diagnostics

Clinical Urgency of Sepsis

- Approximately 49 million sepsis cases globally (2020) with ~11 million deaths.
- Early detection and treatment are critical; delays increase mortality.
- Sepsis is a major contributor to hospital fatalities and ICU burden.



Limitations of Current Diagnostics

- Blood culture: slow (24–72 hours) and insensitive when antibiotics were given.
- Host markers (CRP, PCT) are non-specific and can be elevated in non-bacterial states .
- Clinical scores (qSOFA/SIRS): supportive, not definitive.
- Molecular assays (PCR) require lab infrastructure and known targets.
- Empirical broad-spectrum antibiotic use is common, driving antimicrobial resistance.

Why Bacterial-Specific Biomarkers?

- Early diagnosis and improved clinical outcomes .
- Directly indicate presence of bacteria (not just host inflammation).
- Enable Gram classification.Reduce unnecessary broad-spectrum antibiotic use and combat AMR.

<u>Traditional Approach</u>	<u>Bacterial-Specific Approach</u>
Host response detection	Direct bacterial detection
Delayed immune activation	Immediate bacterial identification
Non-specific inflammation	Bacterial specific marker
Cannot guide antibiotics	Enables targeted therapy
Incremental improvement	Paradigm shift

Difference	Gram-Positive	Gram-Negative	Clinical Significance	Reference
Cell Wall Structure	Thick peptidoglycan (20-80 nm)	Thin peptidoglycan + outer membrane	Determines antibiotic penetration and SERS targeting strategy	Silhavy et al. Cold Spring Harb Perspect Biol 2010;2(5):a000414
Peptidoglycan Content	90% of cell wall mass	5-10% of cell wall mass	Major structural target for β -lactam antibiotics and detection	Vollmer et al. FEMS Microbiol Rev 2008;32(2):149-167
Outer Membrane	Absent	Present (barrier function)	Primary resistance mechanism; barrier to drugs and immune system	Nikaido H. Microbiol Mol Biol Rev 2003;67(4):593-656
Lipopolysaccharide (LPS)	Absent	Present (endotoxin component)	Major sepsis mediator; triggers cytokine storm and shock	Raetz & Whitfield. Annu Rev Biochem 2002;71:635-700
Gram Staining	Purple/Violet (retains stain)	Pink/Red (loses stain)	Primary laboratory identification method (95% accuracy)	Gram HC. Fortschr Med 1884;2:185-189
Antibiotic Susceptibility	Generally, more susceptible	More resistant (outer membrane)	Guides empirical antibiotic selection in sepsis management	Delcour AH. Biochim Biophys Acta 2009;1794(5):808-816
Endotoxin Production	No endotoxin	Lipid A endotoxin present	Key pathophysiology difference in sepsis severity and outcomes	Rietschel et al. FASEB J 1994;8(2):217-225
Clinical Presentation	Device infections, pneumonia	UTI, septic shock, HAI	Influences infection source, severity, and treatment approach	Martin et al. N Engl J Med 2003;348(16):1546-1554

Proposed Bacterial Biomarkers panel

- **Universal Bacterial Detection: Peptidoglycan** fragments (N-acetylmuramic acid)

Reason : Present in ALL bacteria, absent in viruses/fungi/parasites . Direct bacterial confirmation. Unique to bacterial peptidoglycan only.

- **Gram-negative: Lipopolysaccharide (LPS)** — lipid A & O-antigen .

Reason : present Only in Gram-negative bacteria, absent in all other organisms. Strong enhancement, endotoxin correlation

- **Gram-positive: Lipoteichoic acid (LTA) and wall teichoic acids (WTA).**

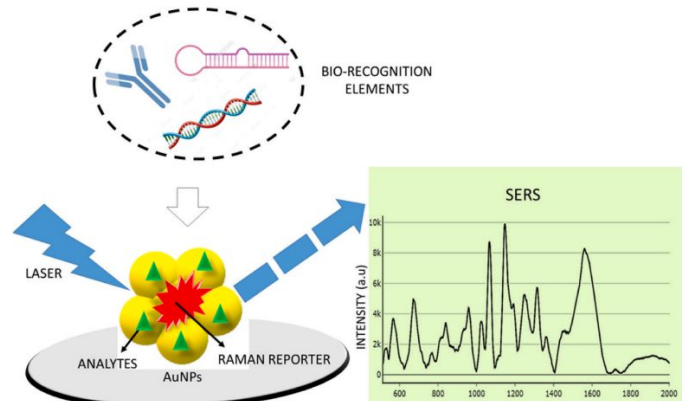
Reason : present Only in Gram-positive bacteria , surface accessibility

SERS Nanobiosensor: Overview

- SERS = Surface-Enhanced Raman Spectroscopy; provides molecular fingerprints.
- Plasmonic nanostructures (Ag/Au) create strong local fields (hotspots).
- Combine SERS substrate with capture probes (aptamers/antibodies) for selectivity.

SERS Fundamentals

- Raman scattering measures vibrational modes; specific peaks = molecule signature.
- Electromagnetic enhancement from plasmon resonance scales..
- Hotspots (nanogaps) give the largest enhancement; single-molecule sensitivity possible.



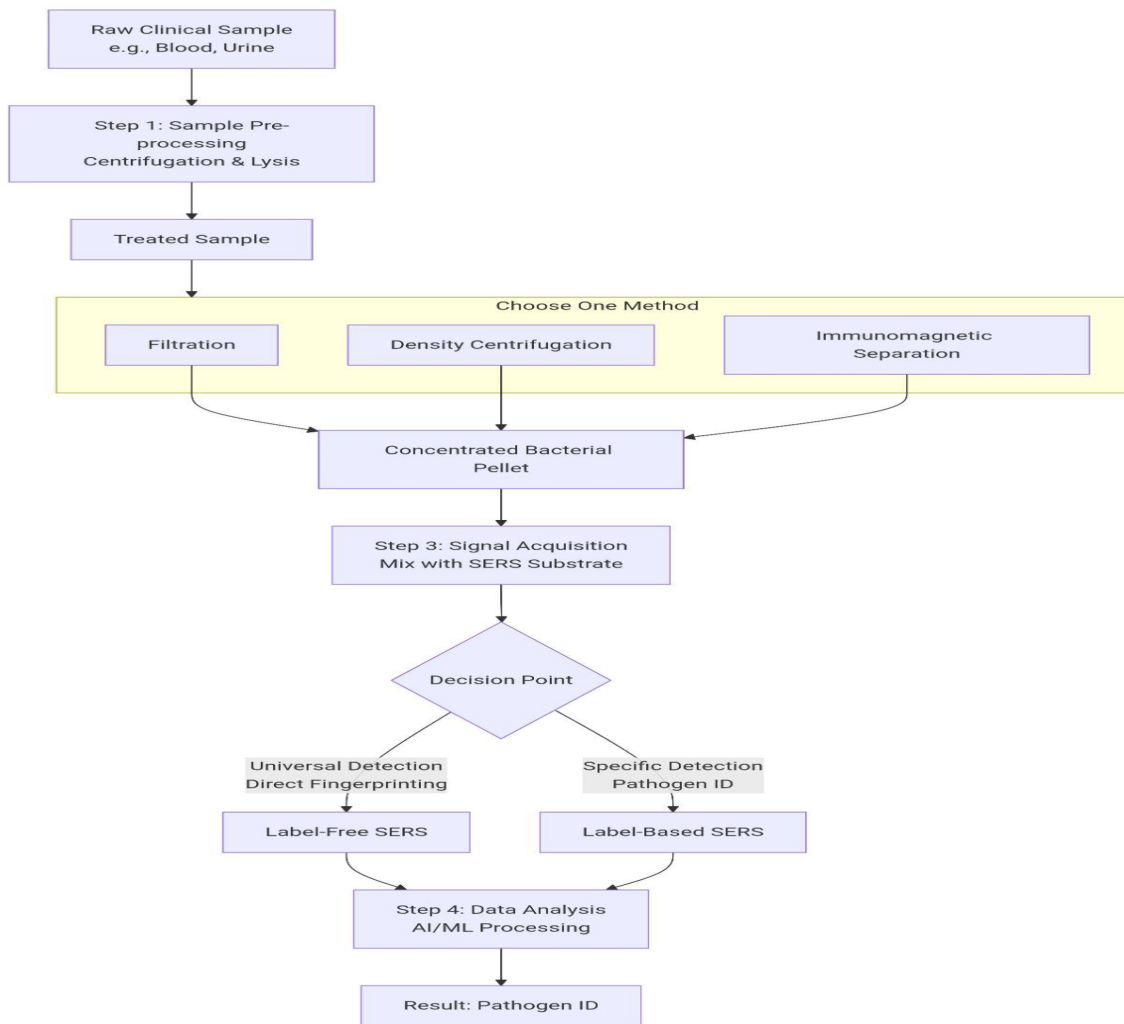
Nanomaterials & Substrate Design

- Choice of metal (Ag stronger SERS; Au more stable) affects resonance.
- Particle geometry (stars, rods, gaps) tunes hotspot intensity.
- Substrate uniformity & reproducible fabrication are essential for clinical use.

Excitation Wavelength & Practical Considerations

- Common lasers: 532 nm (resonant dyes), 633 nm, 785 nm (bio-friendly), 1064 nm (high autofluorescence samples).
- Balance: stronger scattering (shorter λ) vs. fluorescence/background and sample heating.
- Laser power and exposure time must be optimized for clinical samples.

Workflow Pipeline:



Diagnostic Workflow (Patient → Answer)

- 1) Sample collection at bedside/lab.
- 2) Sample prep (optional: concentrate, lyse or microfluidic separation).
- 3) Apply to SERS biosensor (capture probes bind LPS/LTA/PGN).
- 4) Acquire Raman spectrum and analyze for biomarker peaks.
- 5) Output: presence/absence of bacterial markers and Gram classification.

Clinical Utility & Impact

- Rapid Gram-level info enables earlier targeted antibiotics (improves survival).
- Less empiric broad-spectrum use → supports antimicrobial stewardship.
- Portable SERS devices can enable near-patient, point-of-care testing in ED/ICU.

Challenges & Roadmap

- Extensive literature review done and finalization of biomarker .
- SERS simulations initiated to optimize spectral response and substrate design.
- Next: experimental validation of simulation results using standard samples.
- Prototype fabrication and spectral benchmarking planned in next phase.
- **Key challenges:** ensuring signal reproducibility, substrate uniformity, and scaling for real-world use.
- **Long-term goal:** translate optimized SERS workflow into a portable sepsis diagnostic device.

Competitive Advantage & Market Differentiation :

Feature	Blood Culture	PCT/CRP Tests	Our SERS Biosensor
Time-to-Result	24-72 hours	~1 hour	< 30 MINUTES *
Pathogen Specificity	Yes	No	YES (Bacterial & Gram-Type)
AMR Guidance	Delayed	No	IMMEDIATE
Platform	Central Lab	Point-of-Care	POINT-OF-CARE

Key Message: With the help of this project we envision to bridge the gap between speed and specificity

References

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